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9/10/2018

Problem Set 2

1.

a. Shaggy must sit next to Daphne or Velma.
N(Shaggy,Daphne) V N(Shaggy ,Velma)

b. Velma must sit next to Daphne, but cannot sit next to Shaggy or Fred.
N(Velma, Daphne) $\wedge$ -(N(Velma, Shaggy) V N(Velma,Fred))

c. If Daphne doesn't sit next to Fred, then Velma must sit next to Fred.
-N(Daphne, Fred) $\rightarrow$ N(Velma, Fred)

d. Fred must sit next to Daphne if and only if Daphne does not sit next to Shaggy
N(Fred, Daphne) $\leftrightarrow$ -N(Daphne, Shaggy)

2.

a. $\forall m\ \exists n\ (n > m)$ – TRUE
For all m, there exists an n such that n is greater than m
M = 0
N = 6
N > M
6 > 0 : TRUE
(for any value of m, there will be a value number that is bigger than it)

b. $\forall m\ \exists n\ (m > n)$ – FALSE
For all m, there exists an n such that m is greater than n
M = 0
N = 1
M > N
0 > 1: FALSE

c. $\exists n\ \forall m\ ((m \neq n) \rightarrow (m > n))$ - FALSE
There exists an N whereby for every M if M does not equal N then M is greater than N
N = 1
M = 0
0 > 1: FALSE

d. $\forall m\ \forall n\ ((m > 0) \wedge (n > 0)) \rightarrow ((mn > m) \vee (mn > n)))$ - FALSE
If for all natural numbers M and all natural number N,
if M > 0 and N > 0 then M*N > M or M * N > N
M = 1
N = 1
M*N > M
1*1 > 1 FALSE
M*N > N
1*1 > 1 FALSE

3. a.

Step	Result	Justification (Step Used)
1	∀x (A(x) ∨ E(x))	Premise
2	∀x (E(x) → T(x))	Premise
3	A(a) ∨ E(a)	Universal Instantiation (1)
4	E(a) → T(a)	Universal Instantiation (2)
5	E(a)	Hypothesis (4)
6	-A(a)	Disjunctive Syllogism (3 & 5)
7	T(a)	Modus Pollens (4 & 5)
8	-A(a) → E(a)	Logical Equivalences (3)
9	-A(a) → T(a)	Hypothetical Syllogism (4 & 8)
10	-T(a) → - -A(a)	Logical Equivalences (9)
11	-T(a) → A(a)	Double Negative (10)
12	∀x(-T(x) → A(x))	Universal generalization (11)

b.

$\forall x(A(x) \vee E(x))$	For all students, the student is either is a member of the College of Arts and Sciences or a member of the College of Engineering.
$\forall x(E(x) \rightarrow T(x))$	For all students, if the student is a member of the College of Engineering then he/she is required to take Ethics & Technology before graduating
$\forall x(\neg T(x) \rightarrow A(x))$	Therefore: If the student that is not required to take Ethics & Technology before graduating then the student is a member of the College of Arts and Sciences

c.

The statement $\forall x(A(x) \vee E(x))$ can be translated to English as "in all cases (of the defined domain), students are either in the one college or a another" which is not inclusive of one collage and the other, therefore no student can be a member of both colleges.

4. a.

Step	Result	Justification (Step Used)
1	$\forall x(P(x) \rightarrow N(x))$	Premise
2	$\forall x(N(x) \rightarrow (E(x) \vee O(x)))$	Premise
3	$\neg\forall x(P(x) \rightarrow O(x))$	Premise
4	$\exists x\text{-}(P(x) \rightarrow O(x))$	Logical Equivalences (3)
5	$\text{-}(P(a) \rightarrow O(a))$	Existential Instantiation (3)
6	$P(a) \rightarrow N(a)$	Universal Instantiation (2)
7	$N(a) \rightarrow (E(a) \vee O(a))$	Universal Instantiation (2)
8	$P(a)$	Hypothesis (6)
9	$N(a)$	Modus Pollens (6 & 8)
10	$P(a) \wedge \text{-}O(a)$	Logical Equivalence (5)
11	$E(a) \vee O(a)$	Modus Ponens (7 & 9)
12	$\text{-}O(a)$	Simplification (10)
13	$E(a)$	Disjunctive Syllogism (11 & 12)
14	$P(a) \wedge E(a)$	Conjunction (8 & 13)
15	$\exists x(P(x) \wedge E(x))$	Existential Generalization

b.

$\forall x(P(x) \rightarrow N(x))$	For all cases (in the domain), if x is a prime number then n is a natural number
$\forall x(N(x) \rightarrow (E(x) \vee O(x)))$	For all cases (in the domain), if x is a natural number then either x is even or x is odd
$\neg\forall x(P(x) \rightarrow O(x))$	There is a prime number that is not odd

5.

Given that the following is true:

$$\frac{c}{d} x \frac{d}{c} \equiv 1 : \ Inverse\ Law$$

$$\frac{\frac{d}{c}}{\frac{d}{c}} \equiv 1 : \ Multiply\ numerator\ and\ denominator\ by\ d/c$$

Therefore:

$$\frac{\frac{a}{b}\ x\ \frac{d}{c}}{\frac{c}{d}\ x\ \frac{d}{c}} \equiv \frac{\frac{a}{b}\ x\ \frac{d}{c}}{1} \equiv \frac{a}{b}\ x\ \frac{d}{c}$$